

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical and Automation Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical and Automation Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTMPE604C - FINITE ELEMENT METHOD

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The Stiffness matrix is mostly
a) Triangular b) Symmetric c) Unit Matrix d) All of these
- 2 The stiffness matrix is dependant on (i) Geometrical parameters (ii) Material parameters (iii) Type of elements chosen
a) i and iii b) iii c) i, ii and iii d) i
- 3 True or False: More you discretize better is the solution. a) True b) False
- 4 Frictional resistance, Viscous drag & Surface shear are examples of
a) Body force (f) b) Traction force (T) c) Point load (P) d) All of These
- 5
A body for which $\epsilon_z = \gamma_{xz} = \gamma_{yz} = 0$ is said to be in state of _____
a) Plane Stress b) Plane Strain c) Linear strain d) None of the options
- 6 In the finite element method, formulation of the problem results in a system of _____
a) Logical equations b) Algebraic equations c) Arithmetic equations d) None of the options
- 7 Which of The following is Two dimensional Element? (1) Truss element (2) Triangular element (3) Beam element (4) Bar (5) Tetrahedral elements. (6) Rectangular elements (7) Hexahedral elements
a) 3,4 b) 1,3,4 c) 2,6 d) 4,6
- 8 If A and B be real symmetric matrices of size $n \times n$, then
a) $AA^T = I$ b) $(AB)^T = BA$ c) $AB = BA$ d) None of the options
- 9 In Gaussian elimination method, original equations are transformed by using _____
a) Column operations b) Subset Operation c) Mathematical Operations d) Row Operations

- 10 Which of the following matrix multiplication would not be possible
 a) 5×6 matrix with a 6×3 B) 4×3 matrix with a 3×3 c) 2×2 matrix with a 3×3
 d) 5×6 matrix with a 6×1 matrix
- 11 When ever the same shape function is used to express position as well as displacement within the element, it is called as a _____ representation.
 a) Isobaric b) Isochoric c) Isometric d) Natural
- 12 For a bar of length of 20 cm, with left node at $x_1=10$ cm and right node at $x_2=30$ cm; the value of natural coordinate(ξ) at a point $x=20$ cm is
 a) 1 b) 0 c) 0.5 d) -1

Q2. Solve the following.

- A) Write the six major steps involved in displacement formulation of Finite Element Analysis with an illustration. 6
- B) List advantages, disadvantages and applications of finite element method 6

Q3. Solve the following.

- A) Derive the expressions for strain displacement relationships for plane stress condition. 6
- B) Describe plane stress and plane strain conditions? Give examples. 6

Q4. Solve Any Two of the following.

- A) Find solution of following system of equations using gauss elimination approach 6
 $x+2y+3z=14$
 $2x+3y+5z=26$
 $4x+5y+6z=40$
- B) Find the inverse of following matrix. 6

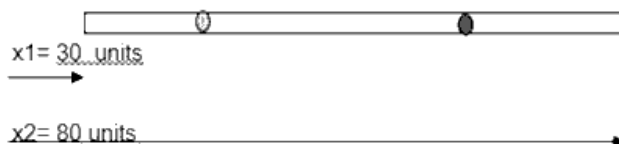
$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

- C) Find eigen values and eigen vectors of the following matrix. 6

$$\begin{bmatrix} 5 & -3 \\ -6 & 2 \end{bmatrix}$$

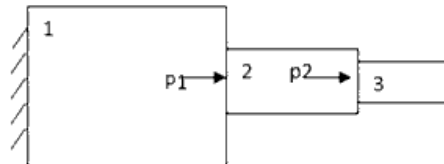
Q5. Solve Any Two of the following.

- A) In reference to the figure; do the following 6
 Evaluate ξ , N_1 and N_2 If $u_1=0.07$ and $u_2= - 0.08$ evaluate value of displacement at $x = 55$ and $x = 75$ units



B) . A structure is loaded as shown below.

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Use following data

Member	E (GPa)	L (mm)	A (mm ²)
1	70	1000	100
2	100	600	60
3	200	600	30

$P_1 = 50 \text{ kN}$; $P_2 = 30 \text{ kN}$

Find the Global Stiffness matrix.

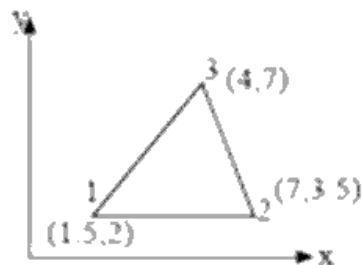
C) In Que 5B find the Stress and Strain in each element.

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Q6. Solve Any Two of the following.

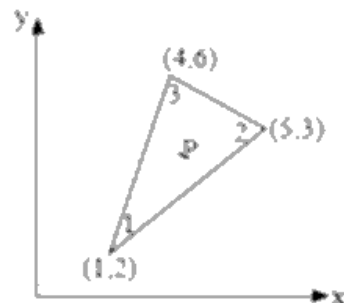
A) Determine the Jacobian of the transformation J for triangular element shown in figure below and further find the area of triangular element.

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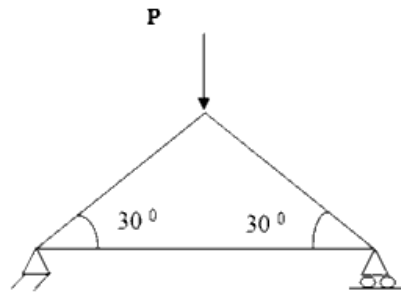
B) The nodal co-ordinates of a triangular element are shown in figure. The x - coordinates of an interior point P is 3.3 and shape function $N_1 = 0.3$. Determine

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N_2 , N_3 and y - coordinate point P.

- C) For the truss given in figure, each isosceles member has length of 500mm, $E=200 \text{ GPa}$, $P=20 \text{ kN}$ and is as shown



Discretise the following truss into finite elements, write element connectivity table, property table and direction cosine table.

*** End ***